Tick-by-Tick Transaction Data, Factor-Based Stock Selection, and LSTM

Data Processing, Factor Evaluation, Turnover-Controlled Backtesting, and Walk-Forward LSTM

Quantitative Trading Project Report

August 9, 2026

Abstract

This project implements a complete research pipeline from tick-by-tick transaction data to daily and minute-level data, 18-factor construction and evaluation, portfolio backtesting under execution constraints, and minute-level LSTM forecasting. The sample contains 300 stocks over 302 trading days. The final low-risk specification preserves the original 15-factor ranking signal and applies a 15% portfolio-volatility target using only information available through the previous trading day, with exposure clipped to the interval $[0.5, 1.0]$. Under next-day-open execution, its full-sample cumulative return is 27.52%, annualized return is 24.95%, annualized volatility is 16.33%, maximum drawdown is -9.58%, and Sharpe ratio is 1.446. In the final 20% holdout period, cumulative return is -4.10%, versus -10.44% for the equal-weight benchmark, for a cumulative active return of 7.08%. Because the holdout period was used to compare strategy versions, it is reported as quasi-out-of-sample rather than strict independent OOS evidence.

Contents

1 Research Objectives and Overall Workflow

This project addresses three questions. First, can tick-by-tick transaction data be transformed without look-ahead bias into reusable daily and minute-level datasets? Second, can 18 cross-sectional factors and an explicit risk budget reduce portfolio volatility and drawdown under realistic trading constraints? Third, can an LSTM provide minute-level ranking information beyond a simple logistic regression in strict out-of-sample time-series validation?

The project follows these steps:

Step 1 Tick-by-tick transaction cleaning

Step 2 Daily and minute-level aggregation

Step 3 Factor construction

Step 4 Factor evaluation

Step 5 Historical-IC weighting

Step 6 Backtesting under execution constraints

Step 7 Robustness analysis

Step 8 Walk-forward LSTM

The report evaluates predictive power, execution costs, and out-of-sample performance separately. All formal strategy statistics use 20-day historical-IC weights that exclude same-day future returns. The version that uses same-day future returns is shown only to demonstrate data leakage and is excluded from the formal conclusions.

2 Data Processing

2.1 Reading, Sorting, and Basic Validation

The tick files are first sorted by stock, trading date, and transaction time. Price, volume, buy/sell direction, and adjustment-factor fields are then checked for valid types and missing values. The project covers 300 stocks and 302 trading days. The minute grid includes the opening call auction, the morning and afternoon continuous-auction sessions, and the 15:00 closing minute, for a total of 253 timestamps per day.

The 11 base output fields are open, high, low, close, volume, number of trades, turnover, aggressive-buy volume, aggressive-sell volume, aggressive-buy turnover, and aggressive-sell turnover. Price fields are stored in daily wide tables and separate intraday wide tables for each date. A 90,600-row daily OHLCV long table is also generated as the unified input to the factor and backtesting programs.

2.2 Price-Unit Recovery and Adjustment

Raw transaction prices are stored as integers scaled by 100, so the actual transaction price is first recovered as

$$P_{i,t}^{\text{real}} = \frac{P_{i,t}^{\text{raw}}}{100}. \tag{2.1}$$

The price fields are then multiplied by the daily adjustment factor $F_{i,t}$:

$$P_{i,t}^{\text{adj}} = P_{i,t}^{\text{real}} \times F_{i,t}. \tag{2.2}$$

Adjusted prices are used for interday returns and technical factors to remove mechanical jumps caused by dividends, bonus issues, and other corporate actions. Turnover continues to be calculated from the actual unadjusted price:

$$A_{i,t} = P_{i,t}^{\text{real}} \times V_{i,t}, \tag{2.3}$$

because turnover represents the monetary value actually traded at the time and should not be altered by the adjustment factor.

2.3 OHLC and Trade-Direction Aggregation

Within each daily or minute-level group, the first transaction price is the open, the maximum is the high, the minimum is the low, and the final transaction price is the close. Volume, number of trades, and turnover are summed. Trade direction is split according to `BSFlag`: 0 denotes an aggressive buy, 1 an aggressive sell, and 2 a call-auction transaction. Aggressive-buy and aggressive-sell volumes and amounts are then aggregated separately.

2.4 Filling Timestamps with No Trades

Prices are forward-filled using only closing prices that have already been observed. Missing daily OHLC values use the previous trading day's close, while missing minute-level OHLC values use the previous minute's close. No future transaction price from the current day is used before the first transaction occurs. Volume, turnover, and number of trades are set to zero when no transaction occurs. This rule prevents look-ahead bias from filling earlier minutes with future prices.

3 Factor Construction and Formulas

Let A denote turnover, V volume, N the number of trades, and O, H, L, C the adjusted open, high, low, and close, respectively. Let $BVol$ and $SVol$ denote aggressive-buy and aggressive-sell volume, and let $r_t^{cc} = C_t/C_{t-1} - 1$. The project constructs 18 factors; the final three directly measure downside risk, idiosyncratic risk, and systematic exposure during market declines.

i) Multi-Window Dispersion of Average Trading Value First, calculate the moving averages of trading value over multiple windows:

$$M_{i,t}^{(n)} = \frac{1}{n}\sum_{j=0}^{n-1} A_{i,t-j}, \qquad n \in \{1, 5, 10, 20\}$$

The factor is defined as:

$$F_{1,i,t} = \ln\left[1 + \mathrm{SD}\left(M_{i,t}^{(1)}, M_{i,t}^{(5)}, M_{i,t}^{(10)}, M_{i,t}^{(20)}\right)\right]$$

ii) 10-Day Active Buy-Sell Imbalance Shock First, calculate the active buy-sell imbalance on each trading day:

$$I_{i,t} = \frac{B_{i,t} - S_{i,t}}{B_{i,t} + S_{i,t}}$$

The factor is defined as the current imbalance minus its average over the previous 10 trading days:

$$F_{2,i,t} = I_{i,t} - \frac{1}{10}\sum_{j=1}^{10} I_{i,t-j}$$

Trades executed during the opening call auction are excluded from active buy and sell volume. The historical average does not include the current trading day, preventing the current shock from entering its own benchmark.

iii) 10-Day Intraday Price Range First, calculate the daily price range:

$$\mathrm{Range}_{i,t} = \frac{H_{i,t} - L_{i,t}}{C_{i,t}}.$$

The factor is the average daily price range over the most recent 10 trading days:

$$F_{3,i,t} = \frac{1}{10}\sum_{j=0}^{9} \mathrm{Range}_{i,t-j}$$

iv) 5-Day Reversal First, calculate the cumulative return over the previous five trading days:

$$R_{i,t}^5 = \frac{C_{i,t}}{C_{i,t-5}} - 1$$

The reversal factor is defined as the negative of this return:

$$F_{4,i,t} = -\left(\frac{C_{i,t}}{C_{i,t-5}} - 1\right)$$

A higher factor value indicates weaker performance over the previous five trading days and therefore a stronger potential reversal signal.

v) 20-Day Abnormal Average Trade Size First, calculate the average trading value per transaction:

$$\mathrm{ATS}_{i,t} = \frac{A_{i,t}}{\max(N_{i,t}, 1)}$$

Apply a logarithmic transformation:

$$X_{i,t} = \ln(1 + \mathrm{ATS}_{i,t})$$

The 20-day moving average is:

$$\overline{X}^{20}_{i,t} = \frac{1}{20}\sum_{j=0}^{19} X_{i,t-j}$$

The final factor is the 20-day rolling standardized value:

$$F_{5,i,t} = \frac{X_{i,t} - \overline{X}^{20}_{i,t}}{s_{20}\left(X_{i,t}, X_{i,t-1}, \ldots, X_{i,t-19}\right)}$$

A larger absolute factor value indicates that the current average trade size is more abnormal relative to its recent historical level.

vi) 10-Day Price-Volume Pressure First, calculate the daily return:

$$r_{i,t} = \frac{C_{i,t}}{C_{i,t-1}} - 1$$

The change in logarithmic trading volume is:

$$\Delta v_{i,t} = \ln(1 + V_{i,t}) - \ln(1 + V_{i,t-1})$$

The daily price-volume pressure is defined as:

$$\mathrm{PVP}_{i,t} = r_{i,t}\Delta v_{i,t}$$

The final factor is the average price-volume pressure over the most recent 10 trading days:

$$F_{6,i,t} = \frac{1}{10}\sum_{j=0}^{9} r_{i,t-j}\Delta v_{i,t-j}$$

A positive value indicates that price and volume changes move in the same direction, whereas a negative value indicates that they move in opposite directions.

vii) 5-Day Overnight-Intraday Return Divergence

The overnight return is defined as:

$$r^{\mathrm{Overnight}}_{i,t} = \frac{O_{i,t}}{C_{i,t-1}} - 1$$

The intraday return is defined as:

$$r^{\mathrm{Intraday}}_{i,t} = \frac{C_{i,t}}{O_{i,t}} - 1$$

The daily overnight-intraday return divergence is:

$$D_{i,t} = r^{\mathrm{Overnight}}_{i,t} - r^{\mathrm{Intraday}}_{i,t}$$

The final factor is the average divergence over the most recent five trading days:

$$F_{7,i,t} = \frac{1}{5} \sum_{j=0}^{4} \left(r_{i,t-j}^{\text{Overnight}} - r_{i,t-j}^{\text{Intraday}} \right)$$

All factors above are initially constructed as raw time-series factors. Before factor evaluation and portfolio construction, each factor is winsorized and standardized cross-sectionally on every trading day:

$$\widetilde{F}_{i,t} = \min \left(\max \left(F_{i,t}, Q_{0.01,t} \right), Q_{0.99,t} \right),$$

$$Z_{i,t} = \frac{\widetilde{F}_{i,t} - \text{Mean}_i \left(\widetilde{F}_{i,t} \right)}{\text{Std}_i \left(\widetilde{F}_{i,t} \right)}$$

Here, $Q_{0.01,t}$ and $Q_{0.99,t}$ denote the 1st and 99th percentiles of the factor's cross-sectional distribution on trading day t, respectively.

viii) 20-Day Price Momentum

$$F_{8,i,t} = \frac{C_{i,t}}{C_{i,t-20}} - 1$$

ix) 20-Day Realized Volatility

$$F_{9,i,t} = \text{SD} \left(r_{i,t-19}, \ldots, r_{i,t} \right)$$

The implementation uses the sample standard deviation with one degree of freedom, i.e., $df = 1$.

x) 20-Day Amihud Illiquidity

$$F_{10,i,t} = \frac{1}{20} \sum_{j=0}^{19} \frac{|r_{i,t-j}|}{A_{i,t-j}}$$

Observations with zero trading value are treated as missing.

xi) 5-Day versus 20-Day Volume Acceleration

$$F_{11,i,t} = \frac{\text{MA}_5(V_{i,t})}{\text{MA}_{20}(V_{i,t})} - 1$$

xii) 10-Day Closing Location Value

The daily closing location value is defined as:

$$\text{CLV}_{i,t} = \frac{2C_{i,t} - H_{i,t} - L_{i,t}}{H_{i,t} - L_{i,t}}$$

The factor is the average closing location value over the most recent 10 trading days:

$$F_{12,i,t} = \frac{1}{10} \sum_{j=0}^{9} \text{CLV}_{i,t-j}$$

When $H_{i,t} = L_{i,t}$, the observation for that trading day is treated as missing.

xiii) 5-Day Overnight Reversal

$$F_{13,i,t} = -\frac{1}{5}\sum_{j=0}^{4}\left(\frac{O_{i,t-j}}{C_{i,t-j-1}} - 1\right)$$

xiv) 10-Day Price-Volume Trend

$$F_{14,i,t} = \frac{\sum_{j=0}^{9} \operatorname{sign}(r_{i,t-j})V_{i,t-j}}{\sum_{j=0}^{9} V_{i,t-j}}$$

Trading volume on positive-return days is assigned a positive sign, while trading volume on negative-return days is assigned a negative sign.

xv) 20-Day Downside Volatility Share

$$F_{15,i,t} = \sqrt{\frac{\frac{1}{20}\sum_{j=0}^{19} \min(r_{i,t-j}, 0)^2}{\frac{1}{20}\sum_{j=0}^{19} r_{i,t-j}^2}}$$

A higher value indicates that negative returns account for a larger proportion of the stock's total realized volatility.

xvi) 20-Day Downside Volatility

$$F_{16,i,t} = -\sqrt{\frac{1}{20}\sum_{j=0}^{19} \min(r_{i,t-j}, 0)^2}.$$

The negative sign ensures that a higher factor value represents lower historical downside risk.

xvii) 20-Day Idiosyncratic Volatility

The daily equal-weighted return of the 300 stocks, denoted by $r_{m,t}$, is used as the market proxy. The code computes the closed-form residual variance of a single-factor market model from rolling 20-day variances and covariances:

$$\sigma^2_{\varepsilon,i,t} = \operatorname{Var}_{20}(r_i) - \frac{\operatorname{Cov}_{20}(r_i, r_m)^2}{\operatorname{Var}_{20}(r_m)}, \qquad F_{17,i,t} = -\sqrt{\max(\sigma^2_{\varepsilon,i,t}, 0)}. \tag{3.1}$$

The negative sign means that a higher factor value represents less stock-specific volatility unexplained by common market movements.

xviii) 60-Day Downside Beta

Only trading days with $r_{m,t} < 0$ are retained, and at least 20 valid observations are required:

$$F_{18,i,t} = -\frac{\operatorname{Cov}_{60}\left(r_{i,t}, r_{m,t} \mid r_{m,t} < 0\right)}{\operatorname{Var}_{60}\left(r_{m,t} \mid r_{m,t} < 0\right)}$$

A higher factor value represents lower systematic risk exposure when the market declines.

3.1 Cross-Sectional Preprocessing

Each factor is winsorized daily at the 1% and 99% quantiles and then standardized cross-sectionally:

$$z_{i,t} = \frac{x_{i,t} - \bar{x}_t}{s_t}. \tag{3.2}$$

4 Factor Evaluation and Interpretation of Figures

4.1 IC, Rank IC, and Portfolio Returns

The evaluation target is the next trading day's close-to-close return. Ordinary IC is the Pearson correlation between the current day's factor values and next-day returns, while Rank IC is the Spearman correlation between the same variables. The report also calculates the mean, volatility, and ICIR:

$$\mathrm{ICIR} = \frac{\overline{\mathrm{IC}}}{\sigma(\mathrm{IC})}. \tag{4.1}$$

In the quintile test, stocks are sorted each day from low to high factor values and assigned to Q1–Q5. Average returns, cumulative net values, and the Q5–Q1 long–short return are then compared across groups. If a factor has negative predictive direction, the composite strategy automatically reverses its sign according to historical IC.

Table 4.1: Summary of factor evaluation

Factor	IC	Rank IC	Rank ICIR	Q5–Q1 cumulative return
Multi-window dispersion of average turnover	-0.0774	-0.1070	-11.711	-75.86%
5-day aggressive buy–sell imbalance	0.0016	-0.0017	-0.266	3.03%
10-day intraday range	-0.0396	-0.0783	-6.871	-41.95%
5-day reversal	0.0303	0.0558	6.357	50.53%
20-day abnormal average trade size	-0.0225	-0.0500	-8.536	-36.25%
10-day price–volume pressure	-0.0424	-0.0752	-10.354	-51.65%
5-day overnight–intraday return divergence	0.0324	0.0539	7.187	80.41%
20-day downside volatility	0.0248	0.0427	4.088	38.78%
20-day idiosyncratic volatility	0.0438	0.0819	9.109	83.44%
60-day downside beta	-0.0127	-0.0062	-0.743	-13.41%

Each figure is read in the same way: the upper-left panel shows Q1–Q5 average returns and is used to check monotonicity; the upper-right panel shows cumulative net value by group; the lower-left panel shows the Q5–Q1 long–short net value; and the lower-right panel shows rolling Rank IC to reveal time variation in factor effectiveness.

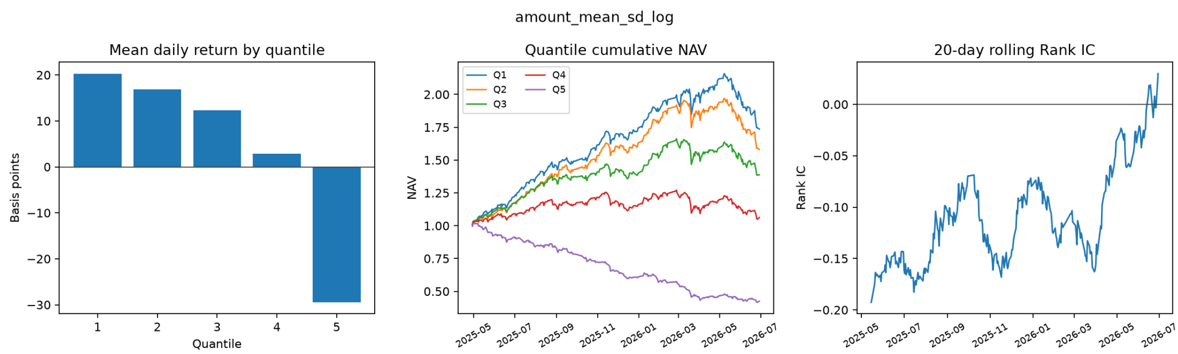

Figure 4.1: Diagnostics for the multi-window dispersion of average turnover. The negative Rank IC and declining portfolio returns indicate that this factor should be used with its sign reversed.

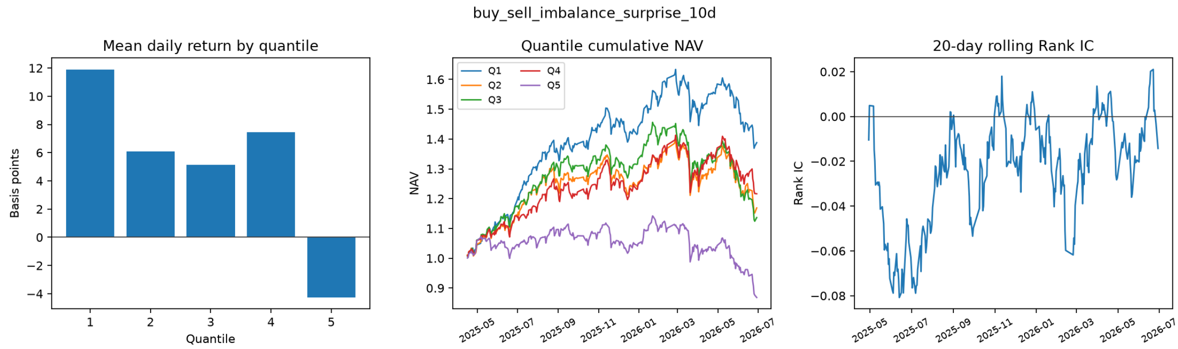

Figure 4.2: Diagnostics for aggressive buy–sell imbalance. Its Rank IC is close to zero and its standalone predictive power is weak, so it should not determine the portfolio direction by itself.

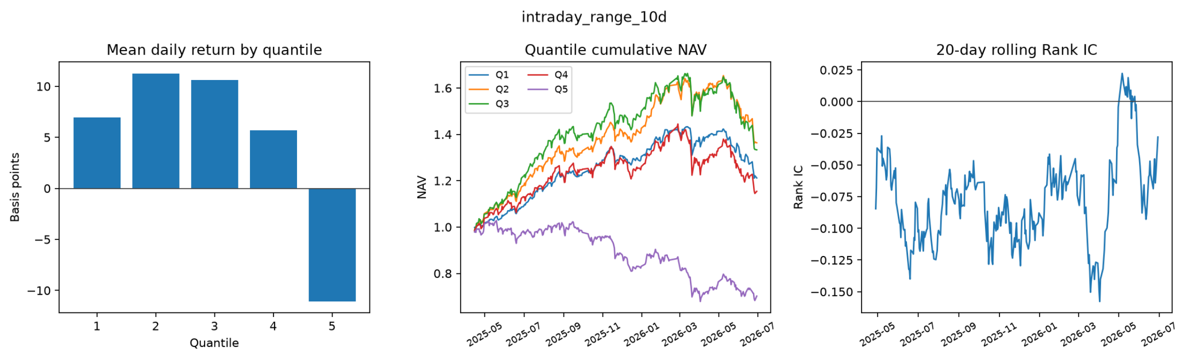

Figure 4.3: Diagnostics for the 10-day intraday range. High-range groups have weaker subsequent returns, making this a stable negative factor.

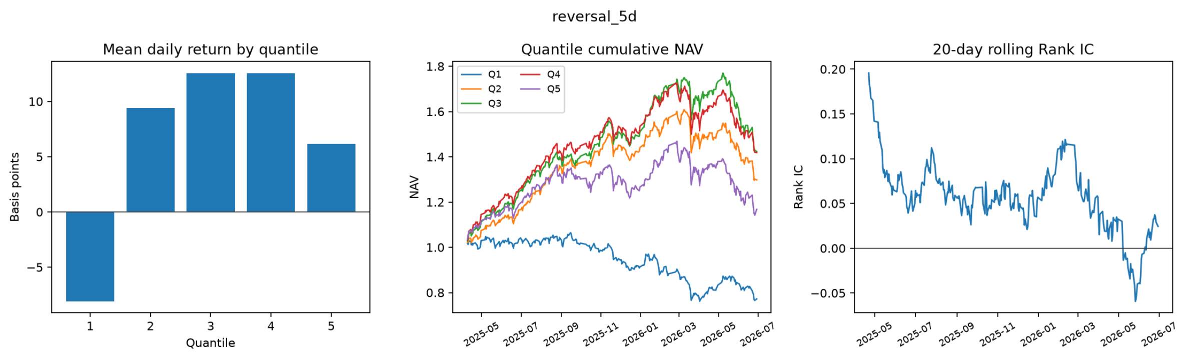

Figure 4.4: Diagnostics for the 5-day reversal factor. Positive Rank IC and positive long–short returns indicate that recently weak stocks experience some subsequent rebound.

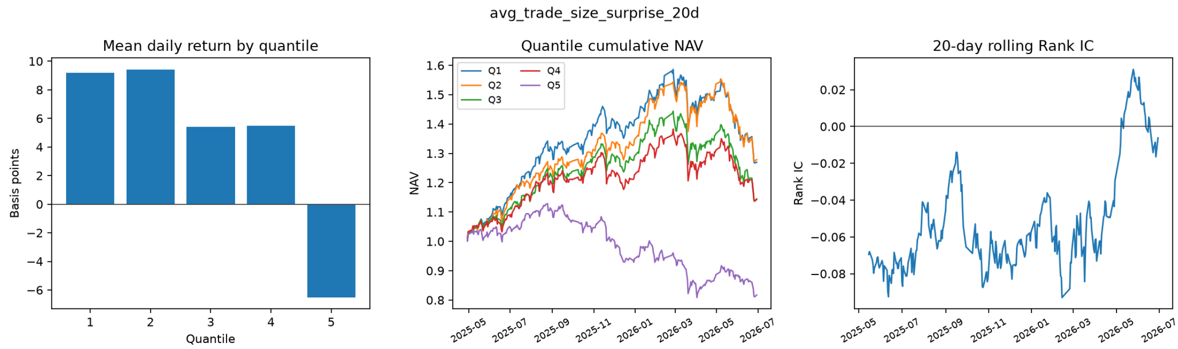

Figure 4.5: Diagnostics for the 20-day abnormal average trade-size factor. Unusually large average trade size is negatively correlated with next-day returns.

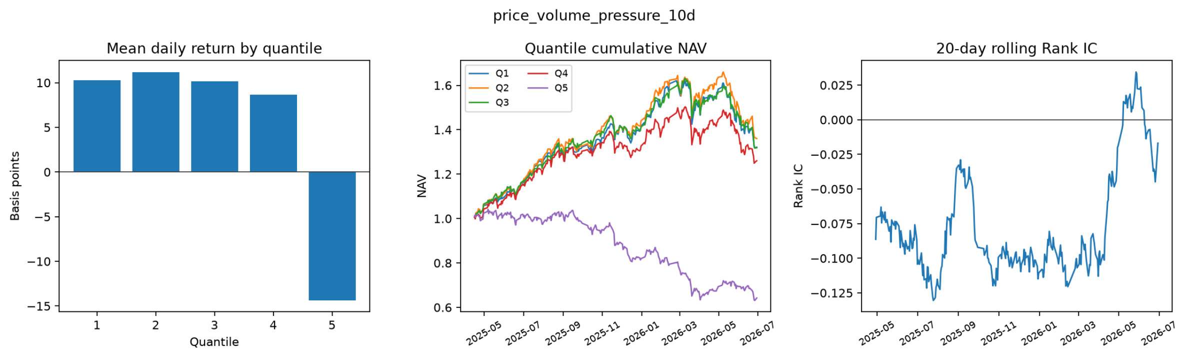

Figure 4.6: Diagnostics for the 10-day price–volume pressure factor. Its predictive direction is negative, suggesting that recent same-direction price and volume pressure may be followed by a correction.

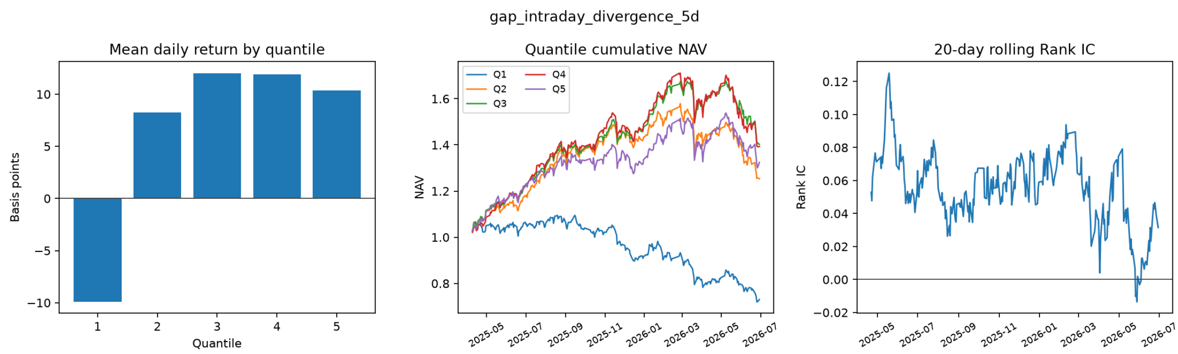

Figure 4.7: Diagnostics for the 5-day overnight–intraday return-divergence factor. Its Q5–Q1 cumulative return is 80.41%, the strongest among the newly added factors.

4.2 Correlation and Long–Short Net Value

Factor correlations are used to identify duplicated information. Among the 18 factors, idiosyncratic volatility and ordinary realized volatility have a correlation of approximately -0.959, showing substantial overlap in their risk information. The new risk factors therefore cannot be added mechanically to the formal signal based on standalone IC. Because the full risk-factor combination worsens drawdown in the holdout period, the final specification retains the original 15-factor ranking and uses only a separate portfolio-volatility target.

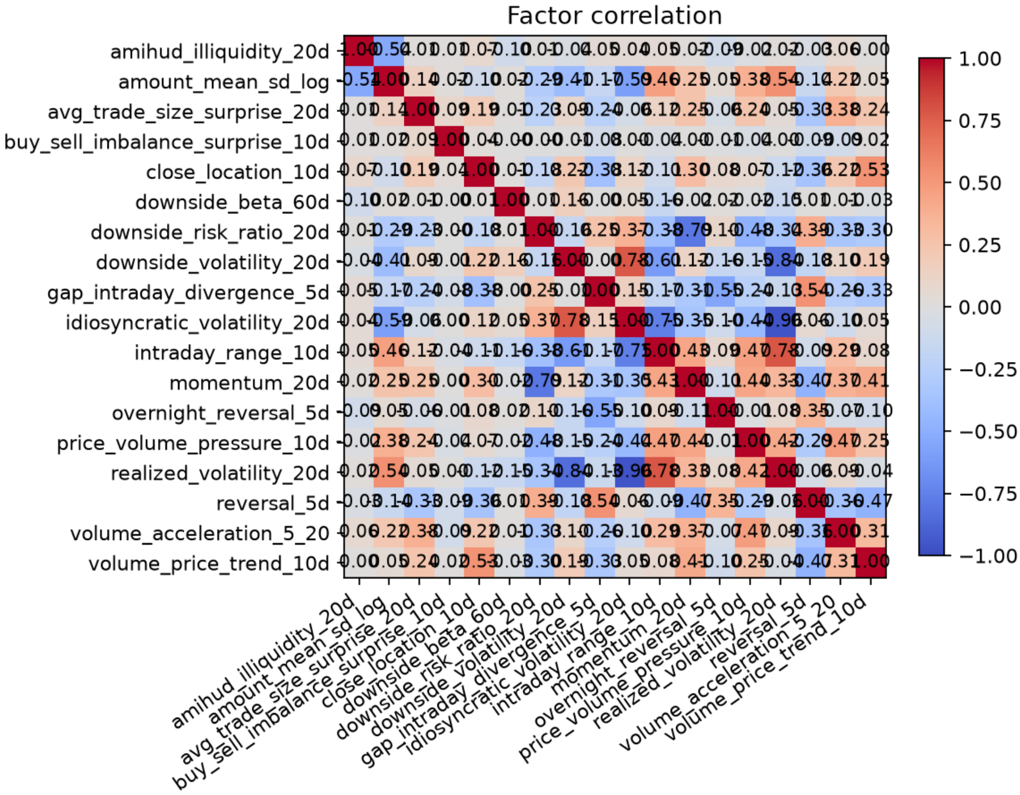

Figure 4.8: Heatmap of cross-sectional correlations among the 18 factors. The new risk factors overlap visibly with conventional volatility measures.

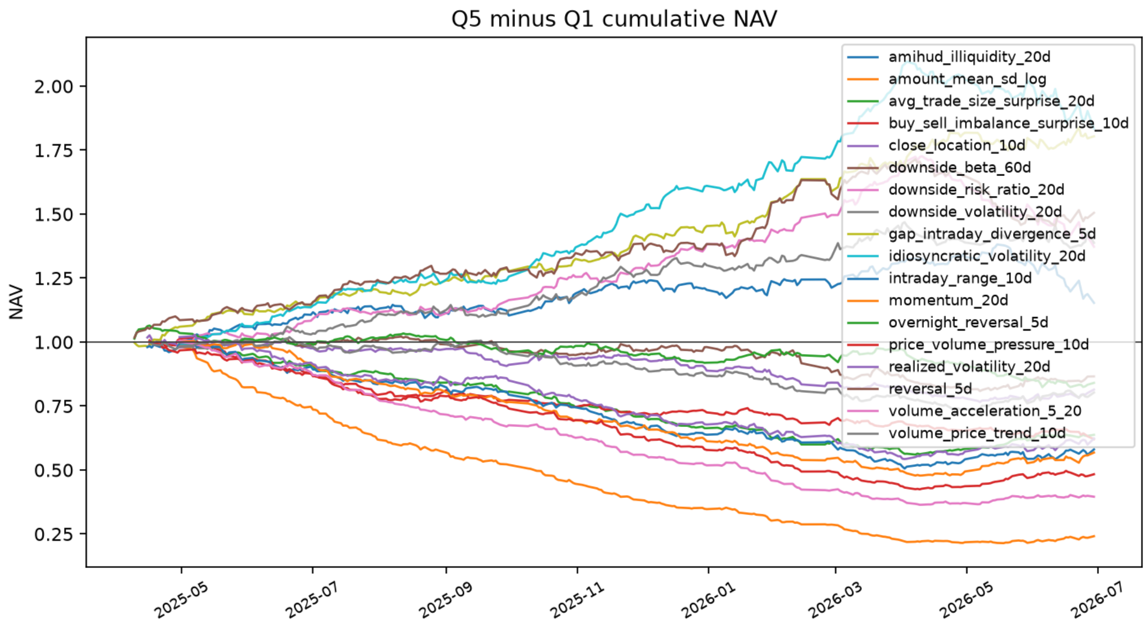

Figure 4.9: Cumulative net values of the 18 factor Q5–Q1 long–short portfolios. Factors with different predictive signs are reoriented according to historical IC before combination.

5 Portfolio Construction and Look-Ahead-Free Backtesting

5.1 Composite Signal Based on Historical-IC Weights

For factor k, the direction and weight on trading day t use only the 20-day historical IC available through $t-1$:

$$\bar{IC}_{k,t} = \frac{1}{20}\sum_{s=t-20}^{t-1} IC_{k,s}, \qquad w_{k,t} = \frac{\bar{IC}_{k,t}}{\sum_j |\bar{IC}_{j,t}|}. \tag{5.1}$$

The composite score is

$$Score_{i,t} = \sum_k w_{k,t} z_{i,k,t}. \tag{5.2}$$

The quantity $IC_{k,t} = \operatorname{corr}(F_{k,t}, R_{t+1})$ cannot be used directly to determine trades on day t, because R_{t+1} has not yet occurred when orders are submitted.

5.2 Candidate Risk Weights and the Adopted Rule

To reduce risk, the research first tests 60-day exponentially weighted ICIR weights. Every statistic uses only IC observations available through $t-1$:

$$m^{60}_{k,t} = \mathrm{EWMA}_{60}(IC_{k,s},\ s \le t-1), \qquad q^{60}_{k,t} = \frac{m^{60}_{k,t}}{\mathrm{EWSD}_{60}(IC_{k,s},\ s \le t-1)}. \tag{5.3}$$

The latest code also implements 60/120-day directional consensus and weak-signal shrinkage. Define

$$m^{120}_{k,t} = \mathrm{EWMA}_{120}(IC_{k,s},\ s \le t-1), \qquad A_{k,t} = \mathbf{1}\left\{\operatorname{sign}(m^{60}_{k,t}) = \operatorname{sign}(m^{120}_{k,t})\right\}. \tag{5.4}$$

The unnormalized and final factor weights are

$$u_{k,t} = A_{k,t} q^{60}_{k,t} \frac{|q^{60}_{k,t}|}{|q^{60}_{k,t}|+1}, \qquad w^{risk}_{k,t} = \operatorname{clip}\left(\frac{u_{k,t}}{\sum_j |u_{j,t}|}, -0.25, 0.25\right). \tag{5.5}$$

A factor is therefore assigned zero weight when its 60-day and 120-day IC directions conflict, while directionally consistent but weak ICIR estimates are continuously shrunk.

Candidate stock weights divide positive composite scores by each stock's prior 20-day volatility:

$$\widetilde{w}_{i,t} \propto \frac{\mathrm{PositiveScore}_{i,t}}{\widehat{\sigma}^{20}_{i,t}}, \qquad \widetilde{w}_{i,t} \le 4\%. \tag{5.6}$$

A position frozen by suspension or a price limit may temporarily exceed 4%. Earlier decomposition results show that replacing the full factor and IC-weighting system worsens holdout drawdown, so the currently adopted specification still preserves the original 15-factor ranking.

The adopted portfolio-volatility target uses gross returns realized before day t:

$$\widehat{\sigma}^p_{t-1} = \mathrm{SD}(r^p_{t-20}, \ldots, r^p_{t-1})\sqrt{252}, \qquad L^{15}_t = \operatorname{clip}\left(\frac{15\%}{\widehat{\sigma}^p_{t-1}}, 0.5, 1.0\right). \tag{5.7}$$

When fewer than 20 observations are available, $L^{15}_t = 1$. An aggressive candidate is

$$L^{18}_t = \operatorname{clip}\left(\frac{18\%}{\widehat{\sigma}^p_{t-1}}, 0.5, 1.1\right). \tag{5.8}$$

The latest code additionally implements tiered drawdown control. Let

$$D_t = \frac{NAV_t}{\max_{s\le t} NAV_s} - 1, \qquad g(D_t) = \begin{cases} 1.00, & D_t > -5\%, \\ 0.75, & -8\% < D_t \le -5\%, \\ 0.50, & -10\% < D_t \le -8\%, \\ 0.25, & D_t \le -10\%. \end{cases} \tag{5.9}$$

Risk cuts take effect on the next tradable session. Recovery requires five consecutive observations of improving drawdown and moves upward by only one tier at a time.

Four observable flags define the market-regime filter:

$$I_t^{trend} = \mathbf{1}\{NAV_t^m < MA_{120}(NAV^m)_t\}, \tag{5.10}$$

$$I_t^{vol} = \mathbf{1}\{\sigma_t^{m,20} > Q_{0.8}^{hist}(\sigma^{m,20})_{t-1}\}, \tag{5.11}$$

$$I_t^{breadth} = \mathbf{1}\left\{ \frac{\sum_i \mathbf{1}(r_{i,t} > 0)}{\sum_i \mathbf{1}(r_{i,t} \text{ valid})} < 0.40 \right\}, \tag{5.12}$$

$$I_t^{corr} = \mathbf{1}\{\bar\rho_t^{20} > Q_{0.8}^{hist}(\bar\rho^{20})_{t-1}\}. \tag{5.13}$$

The historical quantiles use a rolling window of at most 252 days and are lagged by one period. The regime exposure cap is

$$C_t = \begin{cases} 0.60, & I_t^{trend} + I_t^{vol} + I_t^{breadth} + I_t^{corr} \ge 2, \\ 1.00, & \text{otherwise}, \end{cases} \qquad L_t^{final} = \min\{L_t g(D_t), C_t\}. \tag{5.14}$$

A Top-40 portfolio with a Top-80 exit buffer is also run as a separate candidate. These new candidates must be fully rerun and compared on both the research period and the final 20% holdout period. Until those results exist, the current performance tables and adopted-strategy conclusion remain unchanged.

5.3 Turnover Control and Holding Rules

The portfolio holds 30 stocks and rebalances every five trading days. A current holding is retained as long as it remains in the top 60 of the composite ranking; the portfolio is then replenished to 30 stocks from the latest ranking. This buffer prevents small changes between ranks 30 and 31 from automatically triggering trades.

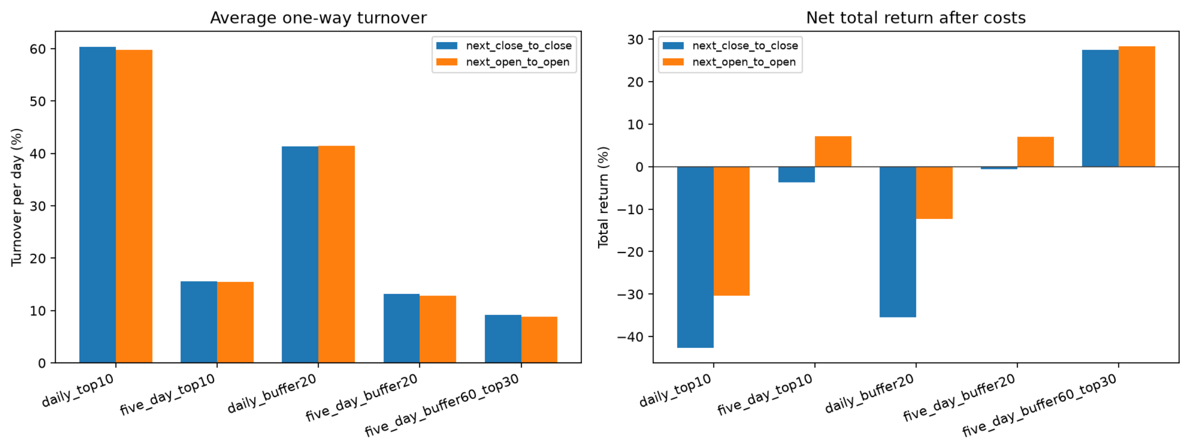

Figure 5.1: Comparison of turnover-control policies. Under next-day-open execution, average daily one-way turnover is 8.81%.

5.4 Execution Timing, Constraints, and Transaction Costs

A signal is formed on day t and can only be executed at the open or close of day $t+1$, after which the position is held until the next corresponding price point. Zero volume or turnover indicates a suspension. A stock at a one-price upper limit cannot be purchased, while a stock at a one-price lower limit cannot be sold. Positions that cannot be sold remain in the portfolio.

Transaction costs have three components: a five-basis-point commission on sell value; five basis points of baseline slippage on both buys and sells; and square-root market impact driven by the order value as a fraction of daily turnover:

$$c_{i,t}^{impact} = \min\left(c_{max}, \eta\sqrt{\frac{Q_{i,t}}{ADV_{i,t}}}\right). \tag{5.15}$$

Here, Q is order value, ADV is daily turnover, and η is the impact coefficient. This design assigns higher costs to less liquid stocks and larger orders.

6 Risk–Return Results and Robustness Tests

6.1 Definitions of Performance Metrics

Cumulative return is $\prod_t(1+r_t)-1$; annualized return is $(1+R)^{252/T}-1$; annualized volatility is $\sigma(r)\sqrt{252}$; maximum drawdown is the largest decline in net value relative to its historical peak; the Sharpe ratio is the annualized average return in excess of the risk-free rate divided by annualized volatility; and the information ratio is the annualized mean active return divided by tracking error. The equal-weight benchmark uses the same-day equal-weight return of stocks that can be purchased on the entry date.

Table 6.1: Full-sample risk–return results of the final strategy

Metric	Next-day close	Next-day open
Cumulative return before costs	37.97%	38.33%
Cumulative return after costs	27.52%	28.30%
Annualized return	24.96%	25.66%
Annualized excess return	20.65%	23.02%
Annualized volatility	18.84%	18.50%
Maximum drawdown	-11.93%	-10.90%
Sharpe ratio	1.277	1.327
Information ratio	1.880	1.838
Average daily one-way turnover	9.22%	8.81%
Total transaction costs	951,129	911,060

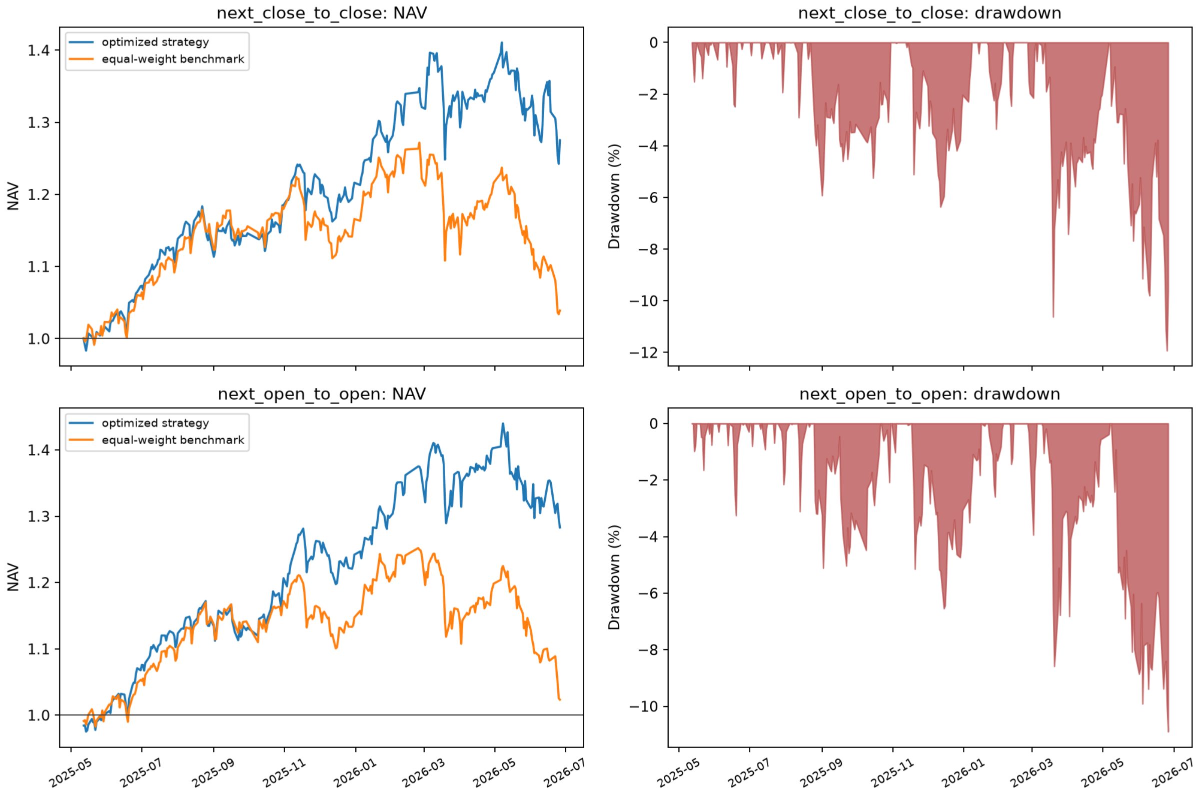

Figure 6.1: Net value of the formal factor strategy, equal-weight benchmark, and drawdown. Next-day-open execution outperforms next-day-close execution in both return and drawdown.

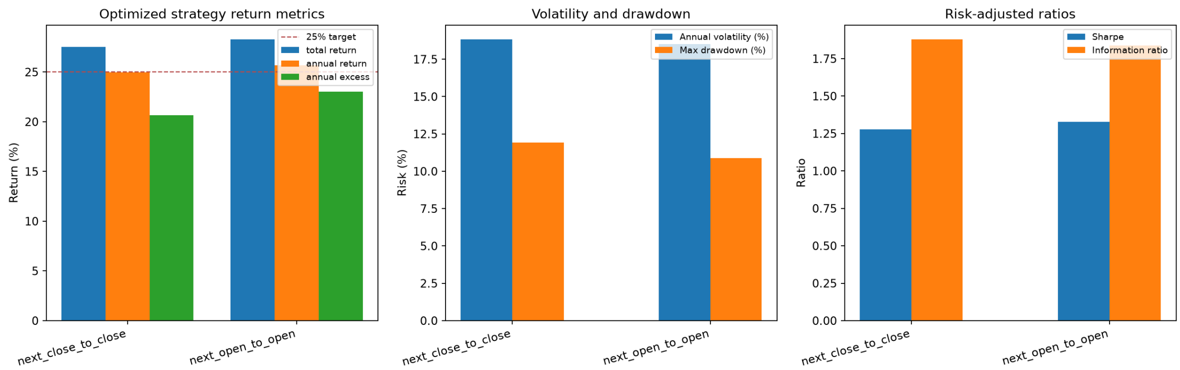

Figure 6.2: Side-by-side comparison of return, volatility, drawdown, Sharpe ratio, information ratio, and turnover.

6.2 15% Volatility Target and Risk Decomposition

The table directly compares the original 15-factor signal at unit exposure, a fixed 1.24-times exposure, and the adopted 15% volatility target. In the full sample, the dynamic target gives up about 0.71 percentage points of annualized return relative to the unit-exposure baseline while reducing annualized volatility from 18.50% to 16.33% and improving maximum drawdown from -10.90% to -9.58%. Its risk improvement is even clearer relative to the fixed 1.24-times exposure.

Table 6.2: Next-day-open risk-budget comparison

Sample	Version	Cumulative Return	Annualized Return	Annualized Volatility	Maximum Drawdown
Full sample	Original signal, 1.00x	28.30%	25.66%	18.50%	-10.90%
Full sample	Fixed 1.24x exposure	33.30%	30.13%	22.94%	-13.60%
Full sample	15% volatility target	27.52%	24.95%	16.33%	-9.58%
Final 20% holdout	Original signal, 1.00x	-5.16%	-21.54%	23.06%	-10.90%
Final 20% holdout	Fixed 1.24x exposure	-6.79%	-27.54%	28.58%	-13.60%
Final 20% holdout	15% volatility target	-4.10%	-17.44%	18.35%	-9.58%

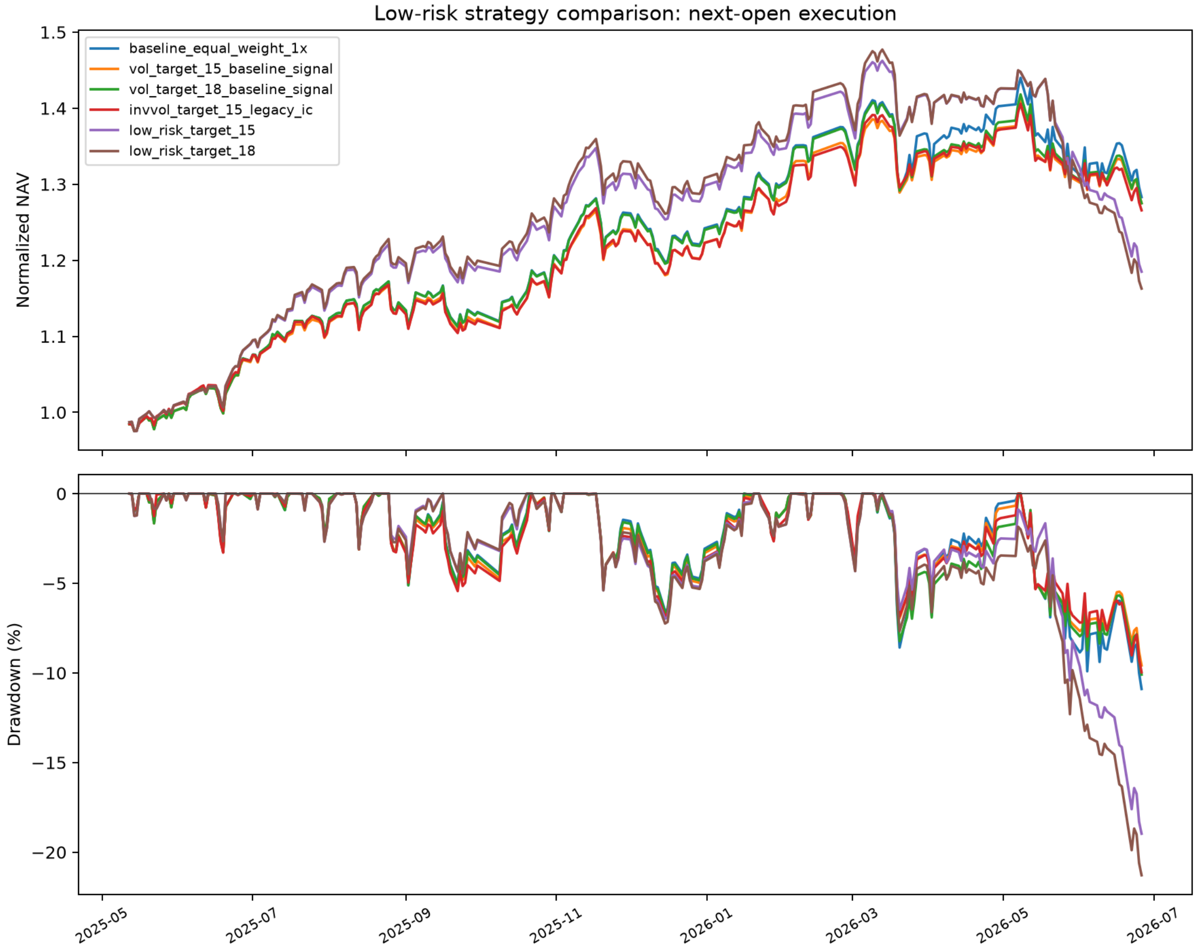

Figure 6.3: Net-value and drawdown decomposition across risk-control versions. Replacing the full factor and IC-weighting system lowers volatility but worsens holdout drawdown; the adopted version preserves the original signal and applies the 15% volatility target.

6.3 Research and Holdout Periods

To distinguish research-selection effects from temporal generalization, the sample is split chronologically into the first 80% as the research period and the final 20% as the holdout period. The adopted 15% volatility-target strategy returns 32.97% in the research period and -4.10% in the holdout period. The equal-weight benchmark returns -10.44% over the holdout period, so cumulative active return is 7.08%. This result indicates relative resilience in a declining market but does not demonstrate stable positive absolute returns.

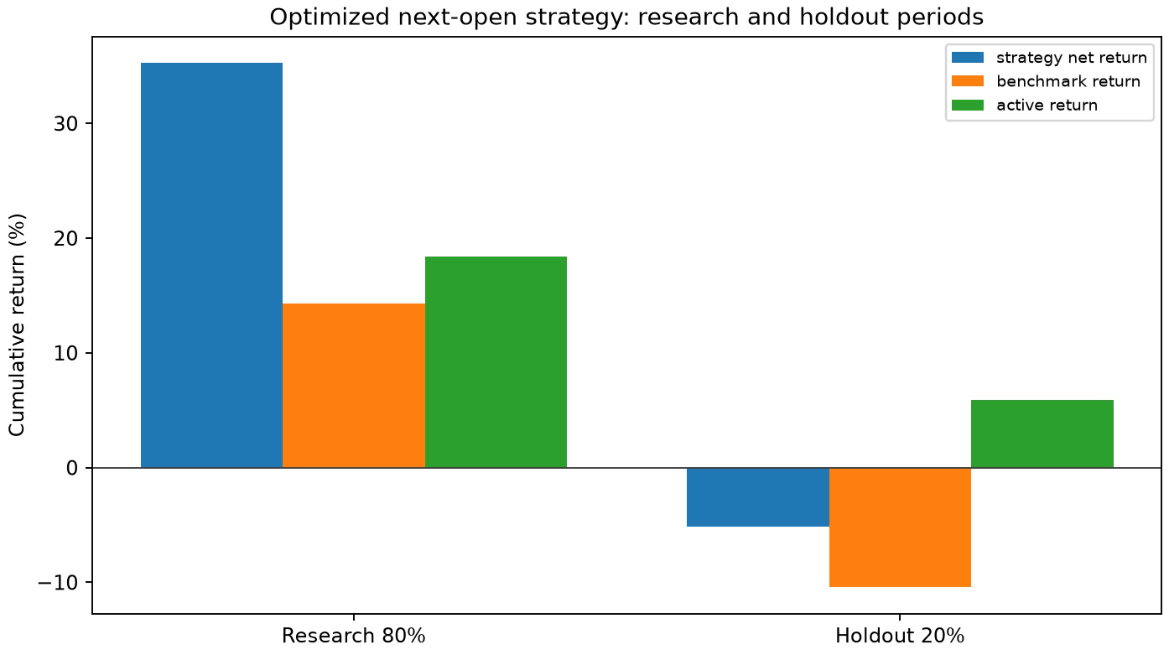

Figure 6.4: Comparison of the full sample, the first 80% research period, and the final 20% holdout period for the adopted 15% volatility-target strategy. The full-sample return of 27.52% is not a substitute for strict out-of-sample validation.

Sample-Period Partition and Definitions

The complete dataset is split chronologically into a research period and a final holdout period. The research period is used for factor exploration, model fitting, parameter selection, and validation; the validation period is therefore part of the research period. Because the final 20% holdout was used to compare risk-control versions in this study, it is quasi-out-of-sample rather than a strictly independent OOS test. "Out-of-sample" describes an evaluation method: the relevant predictions or returns come from data not used to fit the current model or select its parameters. It is not a fixed partition alongside the full sample, research period, validation period, and holdout period. Full-sample results contain both the research and holdout periods and are presented only as an overall backtest, not as independent out-of-sample evidence.

Table 6.3: Unified definitions of sample periods

Term	Definition	Purpose
Full Sample	All dates covered by the report, including the research period and the final holdout period	Present overall net value, returns, drawdown, and transaction costs
Research Period	Interval used for factor exploration, strategy design, model fitting, and parameter selection	Determine factors, number of holdings, rebalance frequency, buffer, and cost parameters
Validation Period	Interval within the research period used for model selection, hyperparameter selection, threshold selection, and early stopping	Compare candidate approaches and determine the final configuration
Holdout Period	Final chronological 20% interval; used here to compare risk-control versions	Provide a quasi-out-of-sample robustness check
Out-of-Sample	A prediction or backtest uses data that were not involved in fitting the current model or selecting its parameters	Describe walk-forward test windows or truly untouched future-period results

Backtest Results by Period

To distinguish research-selection effects from temporal generalization, the backtest results are reported separately for the research period, the final holdout period, and the full sample.

Table 6.4: Backtest results by period

Period	Use of Data	Strategy Cumulative Return	Equal-Weight Benchmark Cumulative Return	Cumulative Active Return	Nature of Result
Research Period (First 80%)	Factor research, parameter selection, and strategy development	32.97%	14.27%	16.36%	Research-period result
Holdout Period (Final 20%)	Risk-control version comparison	-4.10%	-10.44%	7.08%	Quasi-out-of-sample result*
Full Sample	Overall net-value and risk presentation	27.52%	2.34%	24.60%	Full-sample backtest result

* Because the final 20% holdout period was used to compare risk-control versions, this report labels it quasi-out-of-sample rather than strictly independent OOS. Active return is calculated from the strategy net value relative to the benchmark net value; it is not obtained by directly subtracting the two cumulative return percentages.

The adopted 15% volatility-target strategy produces a cumulative return of 32.97% in the research period and an absolute return of -4.10% in the holdout period. The equal-weight benchmark returns -10.44% over the same holdout period. The strategy therefore shows relative resilience in a declining market, but it has not yet demonstrated stable positive absolute returns.

7 Multi-Stock Walk-Forward LSTM

7.1 Sample and Features

The minute-level model uses six stocks and 140 trading days to predict whether the next minute's price will rise from the previous 20 minutes. Input features include log return, minute-level range, log volume, aggressive buy–sell imbalance, and intraday time position. The label is defined as

$$y_{t+1} = \mathbb{I}(C_{t+1} > C_t). \tag{7.1}$$

Each sample contains only information available at or before the prediction time, preventing next-minute data from entering the input window.

7.2 Expanding-Window Validation

The model is a single-layer PyTorch LSTM with a hidden dimension of 16. Validation uses a three-fold expanding-window walk-forward scheme. Each fold contains at least 60 training days, 20 validation days, and 20 test days. Standardization parameters are estimated only from the training period, the validation period is used for model selection, and the three test windows do not overlap. The combined test windows contain 83,880 out-of-sample predictions, of which 31.09% are positive-class observations.

Table 7.1: Out-of-sample classification results

Model	Acc.	Bal. Acc.	Prec.	Recall	F1	ROC AUC	PR AUC
Majority class	68.91%	50.00%	0.00%	0.00%	0.00%	0.5000	0.3109
Logistic regression	68.93%	51.23%	50.51%	4.40%	8.10%	0.6086	0.3993
LSTM	68.98%	52.28%	50.77%	8.10%	13.97%	0.6465	0.4276

7.3 Accuracy and Interpretation

Figure 7.1 presents combined out-of-sample accuracy and three-fold test accuracy in the same figure. The LSTM's overall accuracy is 68.98%, but positive-class observations account for only 31.09% of the sample. A majority-class model that always predicts a decline can therefore achieve 68.91% accuracy. The LSTM improves on the majority-class baseline by only 0.07 percentage points, so 68.98% cannot be interpreted directly as strong directional predictive power.

Balanced accuracy instead calculates recall separately for the positive and negative classes and then averages the two values. The LSTM's balanced accuracy is 52.28%, higher than the majority-class baseline of 50.00% and the logistic regression's 51.23%. This indicates limited but positive incremental identification of minority-class upward observations. Test accuracies for the three folds are 69.22%, 68.88%, and 68.84%, which are stable across time windows but remain very close to the majority-class baseline.

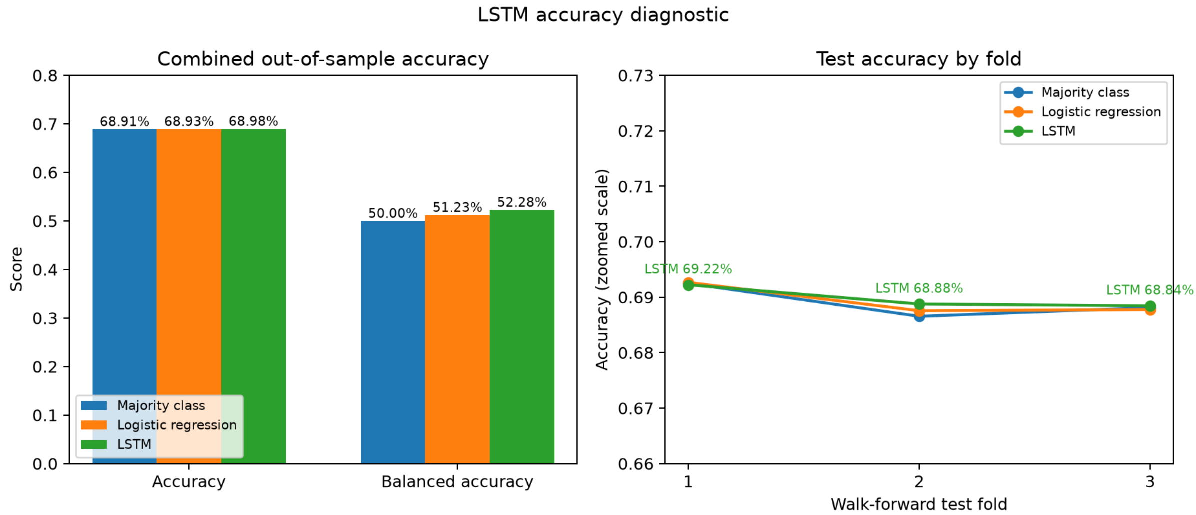

Figure 7.1: LSTM accuracy diagnostics. The left panel compares combined out-of-sample accuracy and balanced accuracy; the right panel compares test accuracy across the three folds. Because of class imbalance, overall accuracy must be interpreted together with the majority-class baseline, balanced accuracy, and ranking metrics.

Combining the positive-class recall of 8.10%, ROC AUC of 0.6465, and PR AUC of 0.4276 yields a more complete conclusion: the LSTM ranks probabilities better than logistic regression, but at the 0.5 threshold it still misses many upward minutes. The model should currently be described as containing some out-of-sample ranking information, rather than as a high-accuracy or directly tradable minute-level strategy.

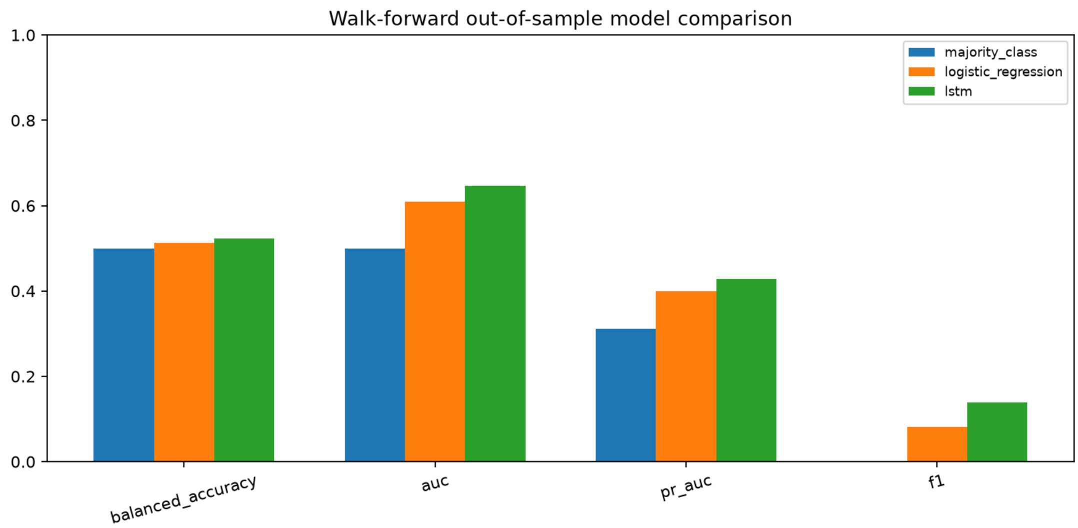

Figure 7.2: Comparison of out-of-sample metrics for the majority-class model, logistic regression, and LSTM. The LSTM has stronger ranking metrics than logistic regression.

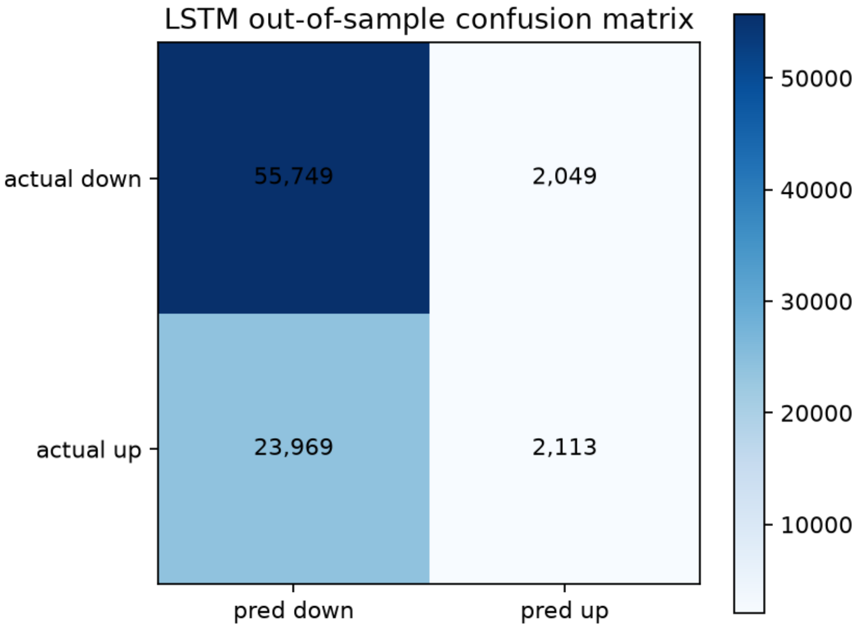

Figure 7.3: Out-of-sample LSTM confusion matrix. There are 2,113 true positives and 23,969 false negatives, indicating low positive-class recall at the 0.5 threshold.

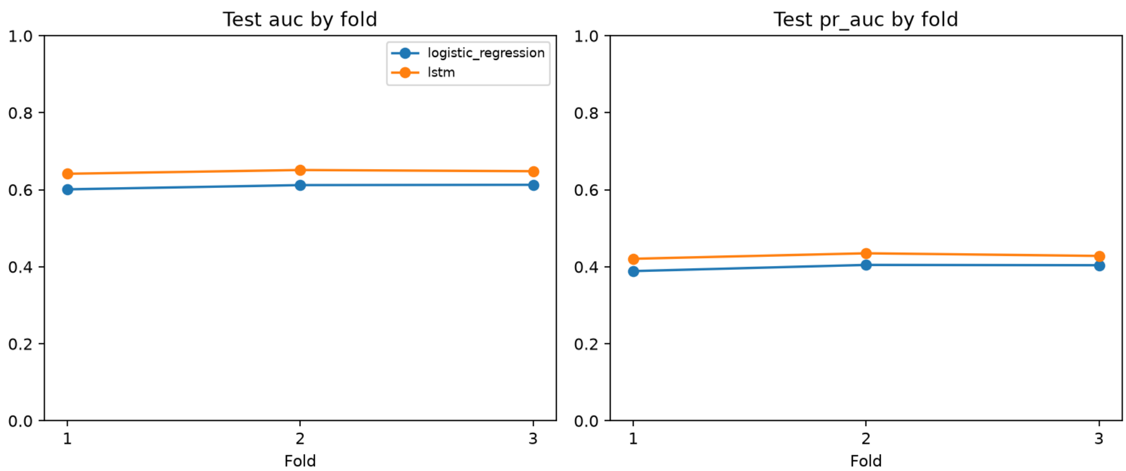

Figure 7.4: Per-fold metrics for the three-fold walk-forward evaluation. Fold-level results reveal whether performance is concentrated in a particular time window.

The LSTM's ROC AUC, PR AUC, and F1 score all exceed those of logistic regression, indicating that the sequence structure provides some additional ranking information. However, overall accuracy is almost identical to the majority-class baseline, and positive-class recall is only 8.10%.

8 Conclusion

- The project constructs 18 daily factors, including downside volatility, idiosyncratic volatility, and downside beta. Their standalone IC can be informative, but their high overlap with conventional volatility factors prevents mechanical inclusion in the formal signal.
- Historical-IC weighting prevents leakage from same-day future returns. The adopted rule preserves the original 15-factor ranking and applies a 15% volatility target with exposure bounded between 0.5 and 1.0.
- Under next-day-open execution, the adopted strategy's full-sample cumulative return after costs is 27.52%, annualized return is 24.95%, annualized volatility is 16.33%, maximum drawdown is -9.58%, and Sharpe ratio is 1.446.
- The final 20% holdout period returns -4.10% versus -10.44% for the equal-weight benchmark. Because this period was used to compare strategy versions, it is quasi-out-of-sample; strict OOS evidence requires a completely untouched future interval.
- The LSTM's out-of-sample ranking metrics exceed those of logistic regression, but its positive-class recall is low at the 0.5 threshold.